Transformations of quadratics

1

a

x	4	5	6	7	8
y	4	1	0	1	4

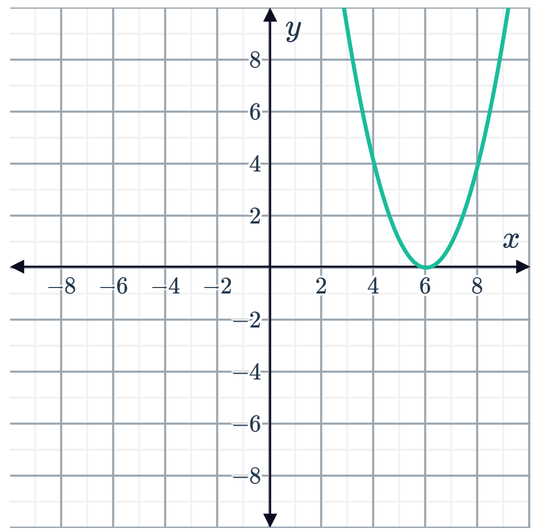
b $y = 0$

c $x = 6$

d $(6, 0)$

e 6 units

f



2

a

x	-8	-7	-6	-5	-4
y	4	1	0	1	4

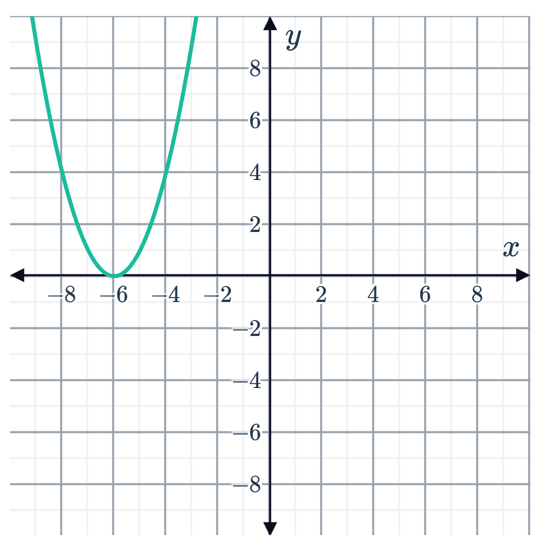
b $y = 0$

c $y = -6$

d $(-6, 0)$

e 6 units

f



3

a $k=-4$

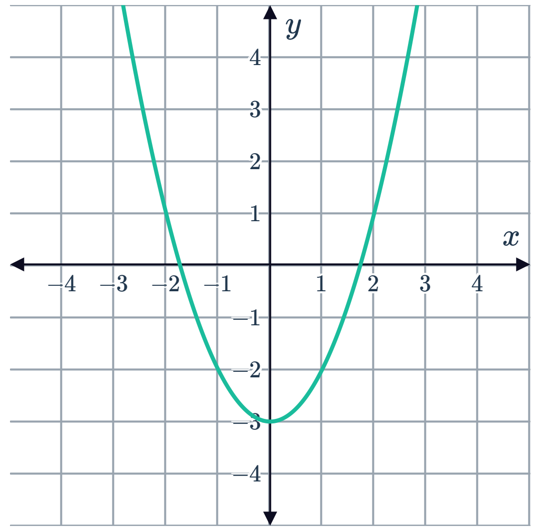
b Horizontal translation of 4 units

4

a

x	-2	-1	0	1	2
y	1	-2	-3	-2	1

b



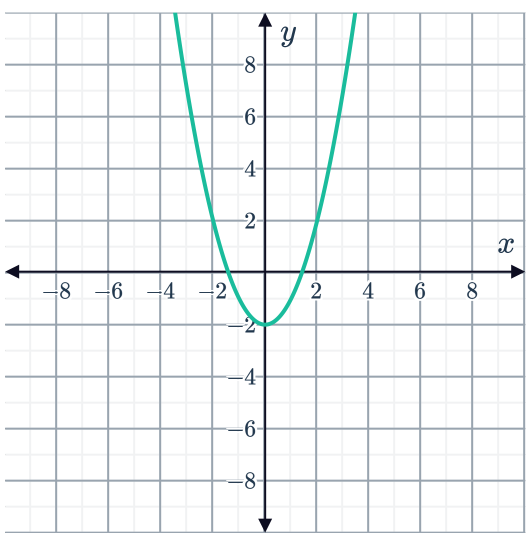
c $y=-3$

d Vertical shift

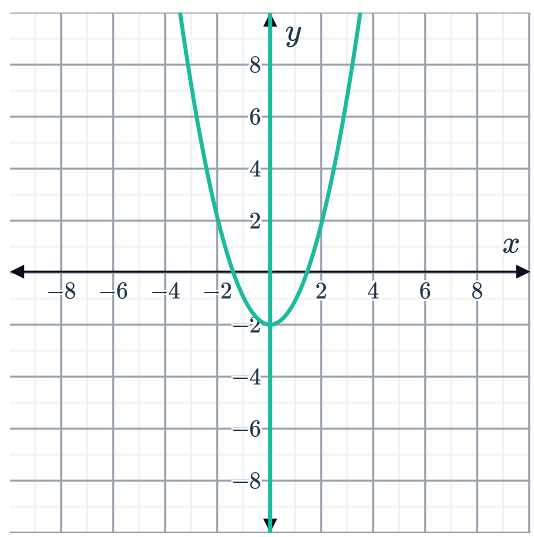
5

a

i



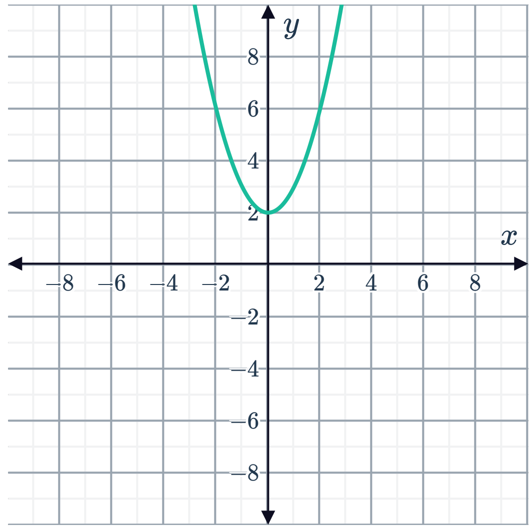
ii



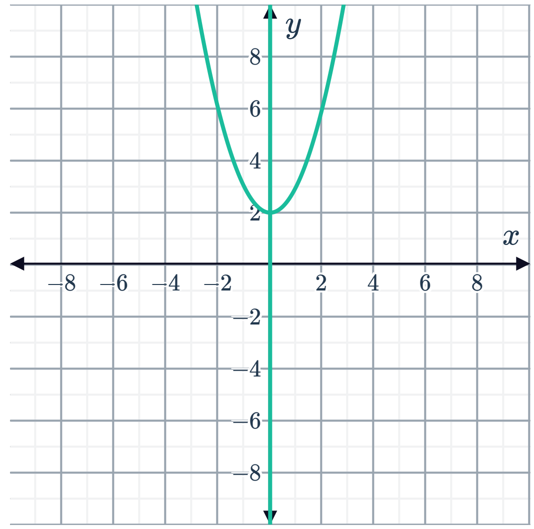
iii $(0,-2)$

b

i



ii



iii $(0, 2)$

6

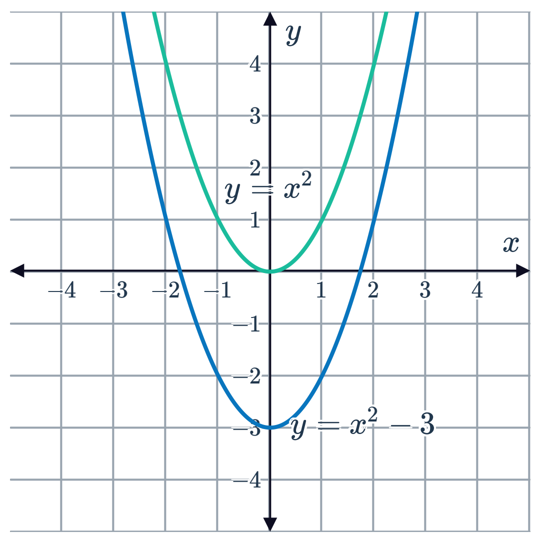
a $y = 3$ **b** $y = -4$ **c** 7 units

7

a

i Move the graph downwards by 3 units.

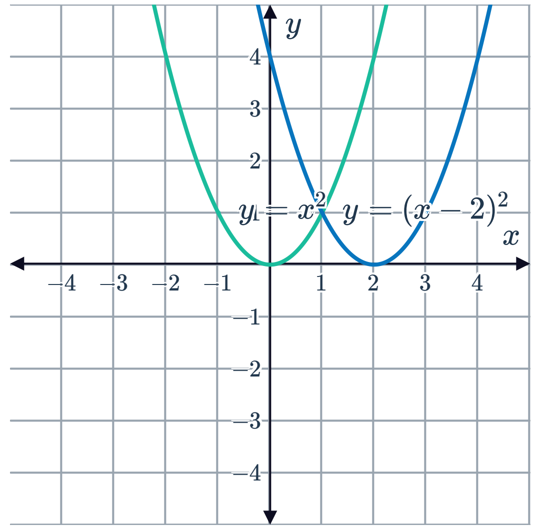
ii



b

i Move the graph to the right by 2 units.

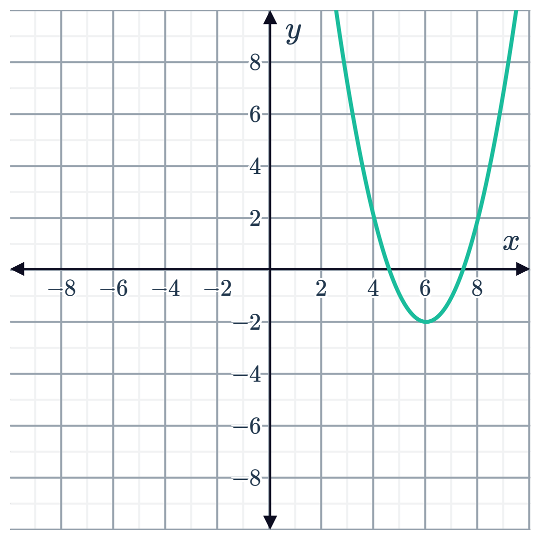
ii



8

a $y = (x-6)^2 - 2$

b



9

a $y = x^2 + 2$

b $y = (x-3)^2 - 3$

10

a

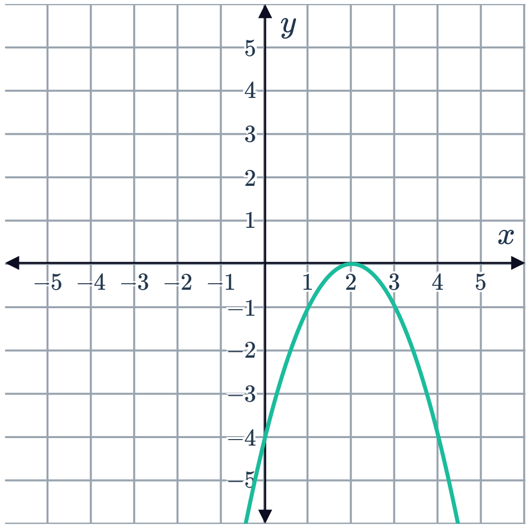
x	0	1	2	3	4
y	-4	-1	0	-1	-4

b $y = 0$

c $x = 2$

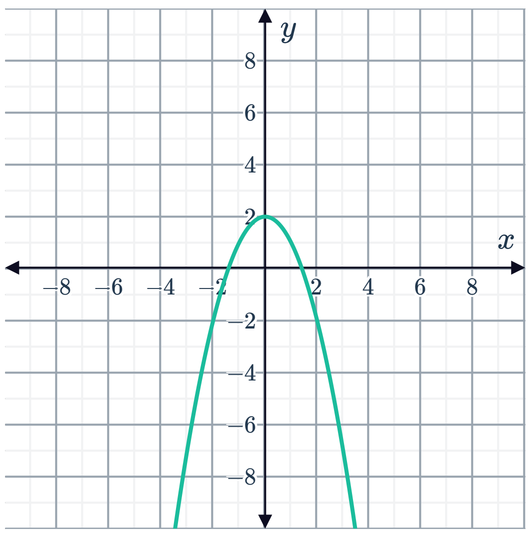
d $(2, 0)$

e



11

a



b $(0, 2)$

12

a No

b No

c No

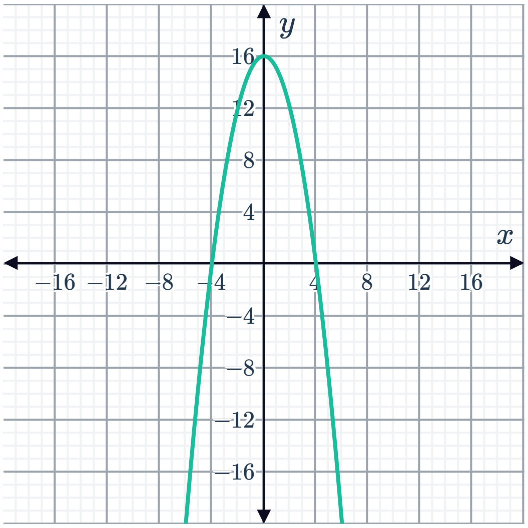
d Yes

13

a $x = \pm 4$

b $y = 16$

c



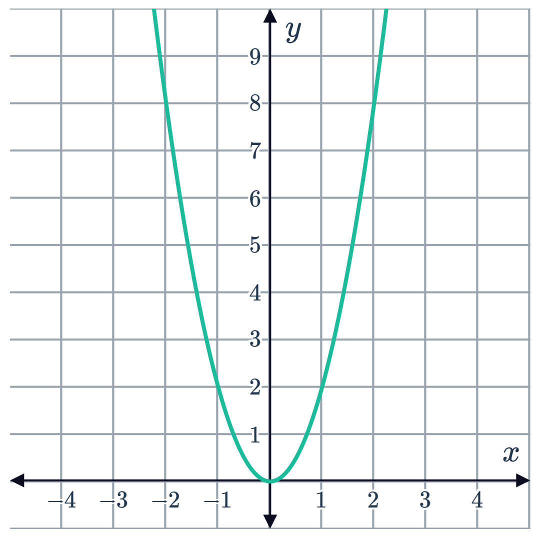
$(0, 16)$

14

a

x	-2	-1	0	1	2
y	8	2	0	2	8

b



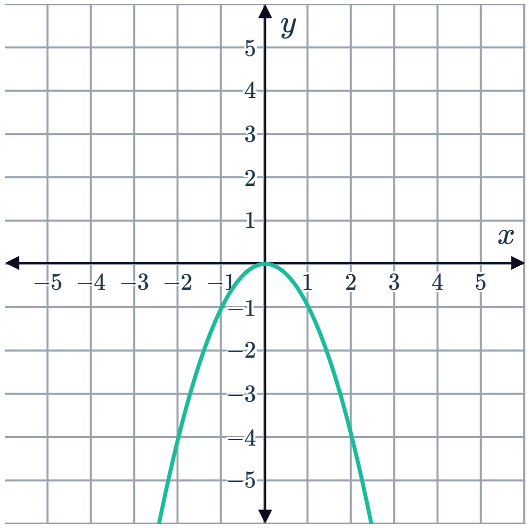
c It narrows the graph.

15

a $a = -1$

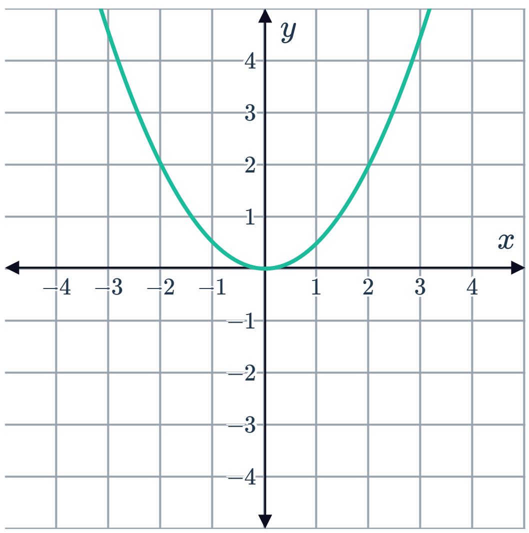
b $(0, 0)$

c



16

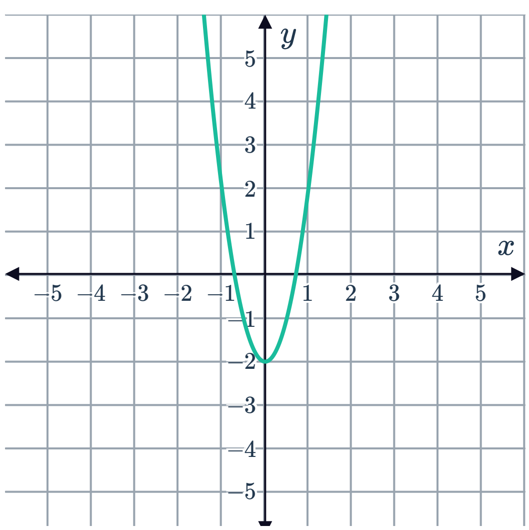
a



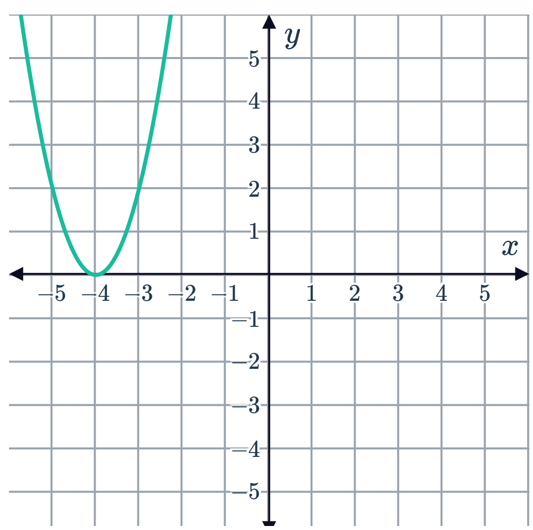
b $y = \frac{1}{2}x^2$

17

a



b



18

a No

b No

c No

19

a 6 units up

b 0 units up or down

c 7 units down

20

a 3 units right

b 7 units left

21

a

i 0 units up or down

ii 0 units left or right

iii No

iv Yes

b

i 2 units down

ii 0 units left or right

iii Yes

iv No

c

i 0 units up or down

ii 4 units right

iii Yes

iv No

d

i 4 units down

ii 3 units right

iii No

iv Yes

22

a $y = 10x^2$

b $y = -5x^2$

c $y = x^2 + 10$

d $y = (x-2)^2$

e $y = -x^2 - 7$

f $y = 8(x+4)^2$

g $y = -(x-6)^2 + 5$

23

a No

b No

c No

d Yes